JULIE BANNWART



PERSONAL INFORMATION

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Date of birth: 12th July 2004
Nationality: French
Pronouns: she/her

EDUCATION

04/2025 –	**PhD student in Mathematics**, Johannes Gutenberg-Universität (JGU), Mainz, Germany. Advisor: Prof. Tom Bachmann.
09/2024 – 03/2025	Semester in JGU, Mainz, Germany, to write my **Master's thesis**: *On the real realization of the motivic spectrum* ko. Advisor: Prof. Tom Bachmann.
09/2023 – 03/2025	**MSc in Mathematics**, Ecole Polytechnique Fédérale de Lausanne (EPFL), Switzerland.
07/2023 – 09/2023	**Summer internship** in the EPFL Laboratory for Topology and Neurosciences ("Summer in the lab" program). Work on N_∞-operads and model structures on poset categories.
09/2020 – 07/2023	**BSc in Mathematics**, EPFL, Switzerland. Thesis: *Model categories and homotopy: the example of topological spaces and simplicial sets*. Advisor: Prof. Jérôme Scherer.
07/2020	**Baccalauréat**, in Forbach, France.

RESEARCH INTERESTS

- Unstable and stable motivic homotopy theory.
- Variants of algebraic K-theory.
- Higher algebra in general.

PREPRINTS

- *The real and complex Betti realizations of motivic multiplicative Thom spectra,* https://arxiv.org/abs/2505.07297, May 2025.

PUBLICATIONS

- *The real Betti realization of very effective Hermitian K-theory*, accepted for publication in J. Inst. Math. Jussieu, https://arxiv.org/abs/2610.20688, May 2025.
- *Realization of saturated transfer systems on cyclic groups of order p^nq^m by linear isometries N_∞-operads*, J. Homotopy Relat. Struct, https://doi.org/10.1007/s40062-025-00377-6, July 2025.
- *When equivariant homotopy theory meets combinatorics* (survey article), Pittsburgh Interdiscip. Math. Rev., vol. 3, pp. 1–27, https://doi.org/10.5195/pimr.2025.56, July 2025.

TALKS

07/2026	Young Topologists Meeting 2026, Copenhagen. "Unstable η-complete and η-periodic motivic homotopy theory »
06/2025	Young Topologists Meeting 2025, Stockholm. "The real Betti realization of very effective Hermitian K-theory, and of motivic Thom spectra"
06/2025	AG Seminar homotopy theory, Regensburg. "The real Betti realization of motivic Thom spectra and of very effective Hermitian K-theory"

TEACHING EXPERIENCE

2026	Exercise sessions for Lineare Algebra 2, JGU.
2026	Exercise sessions for Topology 0, JGU.
2025	Exercise sessions for Algebraic topology II and Foundations of motivic homotopy theory, JGU.
2024	Student assistant for Rings & Fields and Group & Category theory, EPFL.
2023	Student assistant for Linear algebra, EPFL.
2022	Student assistant for Linear algebra, EPFL.
2018-19	Tutoring at high school.